

The Opportunity Cost of Debt Aversion[†]

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We provide evidence of the existence of debt aversion and its negative implications for financial decisions. In a new experimental design where subjects are assigned debt randomly, we quantify the opportunity cost of subjects' debt-biased decisions. One-third of our participants neglect high returns and focus instead on debt repayments. In addition, borrowing to invest is 50 percent less likely when it leads to indebtedness. On average, participants perceive \$1 less in debt as equivalent to \$1.03 in savings. Hence, a debt-averse agent will undertake a 10 percent guaranteed investment only if the cost of borrowing does not exceed 6.80 percent. (JEL C91, D91, G51)

Debt is pervasive in the United States. Whether you buy a house, attend college, or become an entrepreneur, most major investment decisions require access to credit. According to a recent Pew report, eight in ten Americans hold debt, and nearly 70 percent view debt as necessary even if they prefer not to have it.¹ Motivated by its prevalence, the behavioral literature has provided evidence of different mechanisms that can cause people to take on too much debt. For example, present focus (Laibson 1997; Laibson Repetto, and Tobacman 2007; Meier and Sprenger 2010; Ikeda and Kang 2011; Ganong and Noel 2019), self-control motives (Cadena and Keys 2013), cognitive biases (Bertrand and Morse 2011), or naivete about present focus (Allcott et al. 2022) can all cause excessive indebtedness.² In fact, Meier and Sprenger (2010) find present focus to correlate with credit card debt.

However, recent empirical evidence shows instances where the opposite occurs. People do not take on debt when they should and may try too hard to get out of debt instead of investing in higher-return opportunities. One stark example is Berkouwer and Dean (2022), where they show low adoption of a modern charcoal stove in Kenya with an estimated annual return of 300 percent, significantly higher than almost all

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¹Pew Charitable Trusts (2015).

²Ganong et al. (2022) documents failures to consumption smooth in the United States, suggestive of an additional mechanism that could cause too much debt.

local lending rates. In their case, access to credit doubles the willingness to pay for the stove, but it still remains considerably lower than the investment's total financial benefits. Thus, there is a borrowing puzzle. Other examples are Liu and Roth (2022) which documents low uptake of micro-loans by entrepreneurs despite high marginal returns to capital, and Azmat and Macdonald (2020), where they find substantial loss in savings due to “costly” fast home-mortgage repayments in Pakistan. Examples from the education literature also find reluctance to borrow to attend college despite the high returns to education (Callender and Jackson 2005) and impact on career choices for indebted students (Field 2009; Rothstein and Rouse 2011).

While the behavioral literature has primarily focused on present focus as the main force explaining excessive borrowing, it has not been equally successful in providing and testing a mechanism to explain why people are not borrowing enough. In this paper, we aim to fill this gap by showing the ubiquity of debt aversion and its detrimental consequences for investment decisions in a lab experiment. Hence, providing a rationale for why people are reluctant to borrow when high-return opportunities are available. One main advantage of the lab setting, and one of our main contributions, is that it allows us to create a setting where investment returns unequivocally outweigh the cost of borrowing, a feature that is not empirically infrequent³ but has rarely been tested in the literature, reviewed in online Appendix Table A.12.

In our experimental design, subjects own virtual accounts with different interest rates and balances, generating returns over time. During a week-long experiment, they allocate the returns these accounts generate for a total of four allocation decisions. To maximize returns and hence the final payoff from the experiment, subjects should always allocate everything to the account with the highest interest rate. While accounts differ in their interest rates and initial balances, only interest rates matter for maximizing returns. In our first treatment, subjects can allocate points only to accounts that start with positive balances, providing a baseline for what fraction of participants follows the return-maximizing strategy in the absence of debt. Our main treatment variation changes the accounts' initial balances, where two now have a negative starting balance. The lab setting allows us to assign debt randomly while keeping actions and payoff opportunities identical across conditions, where factors like selection or uncertainty play no role.

First, we show in an allocation decision problem that subjects with debt are less likely to maximize returns as they focus on repaying negative balances. Subjects with debt are three times less likely to maximize returns across all decisions than subjects in our baseline group. This difference is not driven by participants behaving randomly or erratically but rather by a somewhat sophisticated strategy of repaying debt first, consistent with a debt-aversion motive. Thirty-eight percent of subjects fully repay at least one debt account, and 20 percent repay both. Once accounts are zeroed out, these participants maximize returns in subsequent decisions. In fact, one out of four participants states using a strategy of first paying debt, then maximizing returns when describing their choices. When outstanding debt becomes larger and harder to repay fully, we find a heterogeneous response. More subjects do not

³ See Berkouwer and Dean (2022) for other examples of high-return investments in the literature.

attempt to repay and maximize returns, but those who try to repay their debt accrue larger financial losses.

In additional treatments, we increase subjects' agency by allowing them to redistribute balances across accounts. If redistribution is also debt-biased, we can see whether debt aversion compounds and causes larger financial losses. This additional option could benefit subjects, allowing them to further increase their returns by consolidating toward the account with the highest interest rate. When subjects can redistribute toward debt accounts, we find that 34 percent of participants partially exhaust their high-interest account toward a low-interest one, dampening their returns. In contrast, when redistribution toward debt accounts is not possible, only 14 percent of the subjects follow such a strategy, and twice as many subjects now increase their payoffs by redistributing towards the highest interest account. By running these additional treatments, we show that redistribution decisions, and not just allocation decisions, are biased toward early debt repayment.

After finding that debt aversion distorts allocation and redistribution decisions, we explore how debt aversion manifests in borrowing decisions. In two additional treatments, we introduce borrowing opportunities and vary whether they incur debt. These borrowing opportunities allow subjects to benefit from the highest interest rate account even more. Nevertheless, only 34 percent of subjects borrow the maximum amount when borrowing involves debt, compared to 63 percent when borrowing does not incur debt. Almost one-half as few. This stark difference persists even among subjects who otherwise maximize returns. Therefore, we show that debt aversion hinders borrowing behavior and can impact people without outstanding debt.

Altogether, our results indicate that many subjects attempt to repay their debt as quickly as possible and are reluctant to borrow in a context with no financial downside. Using our experimental data, we estimate a stylized model where agents overweight debt and find that, on average, participants perceive a dollar less of debt as 1.03–1.07 dollars in savings. At face value, these estimates imply that for a 10 percent guaranteed return, people will borrow only if the interest rate is lower than 6.28 percent or 3.29 percent if we use the higher end of the range. These estimates are robust to using a one-shot version of our experiment instead of the main week-long, which can easily be incorporated into surveys to measure debt aversion in the field.

While it seems at first that our results and main intuition starkly contrast the predominant view in the literature that people take on too much debt, often driven by present focus, we show in a calibration exercise that is not necessarily the case. Using two examples inspired by Berkouwer and Dean (2022) and (Allcott et al. 2022), we find that in situations where debt finances present consumption, present focus is likely to be the main channel, and hence, we might expect excessive borrowing. In fact, most of the literature on excessive borrowing focuses on cases where benefits precede loan repayments, such as credit cards and payday loans, summarized in online Appendix Table A.12. In contrast, for situations where borrowing is used in long-term investment, e.g., a charcoal stove or college, debt aversion becomes more predominant, and the concern is the opposite: people do not borrow enough and under-invest. These examples highlight how maintaining debt

is not inconsistent with debt aversion as long as consumers are also present focus, a point first made by Prelec and Loewenstein (1998).

Models in the behavioral literature have mainly focused on understanding why people might borrow in excess, with present focus as the predominant mechanism. On the other hand, less direct work has been aimed at testing for debt aversion. Most exceptions are from the education literature, focused on college admissions and financial aid. Callender and Jackson (2005) shows that in England, the fear of debt can prevent qualified students from attending college, especially those of more disadvantaged backgrounds. Using a Canadian sample, Eckel et al. (2007) finds no evidence of debt aversion while choosing different types of financial aid. In a US sample, Field (2009) offers two equivalent financial aid packages, one involving a grant and the other a loan, and finds that students who receive loans are less likely to work in public jobs, suggesting the urge for loan repayments might make high-paying jobs in the private sector more appealing.⁴ Caetano, Palacios, and Patrinos (2011) shows, using a Mexican sample and hypothetical decisions, how contracts labeled as debt are less likely to be accepted. Avery and Turner (2012) argue that given the high returns to education, the claim that student borrowing is “too high” can clearly be rejected in most cases. Ultimately, it is unclear whether these borrowing and career decisions are optimal and the role of debt aversion given other factors, such as job uncertainty, future borrowing constraints, or inattention to loan features, to mention a few. Our contribution is to provide a clean and sharp test of debt aversion, in which the cost of borrowing is unequivocally below the benefits of investing. We represent investments as guaranteed returns on a virtual account, showing how debt aversion can be a more general phenomenon explaining under-investment decisions outside of education, such as the adoption of new technologies (Berkouwer and Dean 2022).

Most closely related to our work is the aforementioned Prelec and Loewenstein (1998) seminal paper on debt aversion. They propose a model of mental accounting and timing of enjoyment where people prefer to pay for consumption in advance, allowing them to enjoy future consumption without the mental burden of paying. In their framework, debt is construed as either consuming something before paying for it or getting advance payment for future work. Hence, their key tradeoff is debt versus consumption utility.⁵ Although related, this is fundamentally different from our framework, where we focus on investment decisions in a setting without any consumption. Hence, agents are trading debt versus investment returns, both in monetary terms and without timing differences. This is not a trivial distinction since many of the empirical borrowing puzzles previously discussed refer to investment decisions, and none of our lab decisions involve consumption of any sort.

Our paper contributes to the growing body of evidence showing that some consumers struggle to make debt repayment decisions that minimize accrued interest. Experimental subjects prefer closing out small debt accounts (Amar et al. 2011; Besharat, Carrillat, and Ladik 2014; Beshart, Varki, and Craig 2015) and concentrating repayments instead of spreading them out (Kettle et al. 2016). In recent

⁴Other nonexperimental studies show how student debt can also lead to higher paid jobs (Rothstein and Rouse 2011; Luo and Mongey 2016), deter graduate school enrollment (Fos, Liberman, and Yannelis 2017), and reduce labor search effort (Ji 2016).

⁵Another example of such tradeoff is Meissner (2016), who finds in a consumption/saving experiment that deviations from optimal behavior are higher when subjects have to borrow.

work, Ozyilmaz and Zhang (2019) document how subjects struggle to minimize interest payments in an all-debt environment compared to a nondebt setting. Further evidence on credit card repayments shows that conditional on repaying debt, many borrowers do it suboptimally (Stango and Zinman 2009; Keys, Pope, and Pope 2016; Ponce, Seira, and Zamarripa 2017; Gathergood et al. 2019). In contrast, we find in our setting that those who repay their debts do so optimally, minimizing interest rates. We hypothesize this is a consequence of the transparency of our design, in which interest rates are clearly displayed and quite different from one another. This is in line with previous findings by Ponce, Seira, and Zamarripa (2017), where they show that consumers are insensitive to interest rates when paying their credit cards. However, they are sensitive to an intervention making price reductions more salient, suggesting inattention or lack of information.⁶ We contribute to this literature by showing that even when repayments are based on interest rates, there is the potential for significant foregone opportunities. We expect these to be more substantial in contexts where high-return opportunities are available and acknowledge this is hardly the case for people paying credit card debt or payday loans with hefty interest rates.

Finally, our paper relates to the experimental and psychology literature on scarcity, showing how people in more precarious financial situations make worse financial decisions. Prior work on scarcity has found that lacking money or time leads to poorer decisions (Shah, Mullainathan, and Shafir 2012; Mani et al. 2013), that people under financial strain are less productive (Kaur et al. 2019) and have a higher cognitive load (Haushofer and Fehr 2014; Schilbach, Schofield, and Mullainathan 2016; Ridley et al. 2019), and that wealth impacts the relative perception of assets and debt balances (Sussman and Shafir 2012). Sussman and Shafir (2012) also finds that people with positive net worth are perceived as wealthier when they have lower debt. This is consistent with our evidence, where net wealth is always positive, and a possible micro foundation for why we observe debt aversion. Both Shah, Mullainathan, and Shafir (2012) and Mani et al. (2013) describe attention-driven scarcity as causing individuals to “tunnel” on an urgent item while neglecting potentially important but not urgent items. Such effects might help explain the prevalence of debt traps among the poor (Karlan, Mullainathan, and Roth 2019). In the Shah, Mullainathan, and Shafir (2012) experiment, this causes participants to borrow excessively due to a pressing time constraint. While people in precarious financial situations can potentially be more exposed to indebtedness, it is important to separate debt from poverty. One key feature of our design is the introduction of debt independent of scarcity, which we argue is unlikely to drive our results for two reasons: first, there is no particular urgency to our debt as payments are at the end of the study; and second, it is not clear how attention would cause reluctance to borrow from debt accounts that start with a zero balance.

As with any experimental evidence, further work is needed to assess the relevance of our results outside the lab. Other behavioral measures have been shown to correlate with choices in nonexperimental settings. For example, Meier and Sprenger (2010) show that present focus individuals are more likely to have credit card debt.

⁶See Bertrand and Morse (2011); Burke and Wang (2016) for examples of how lack of information can drive borrowing decisions.

Debt aversion could also become a behavioral trait, with experimental measures of debt aversion having the potential to shed light on financial decisions. To this end, we also develop and test a simplified version of our design that others can easily incorporate into surveys and fieldwork.

The next section describes the experimental design and our main hypotheses. Section II contains our main results on the existence of debt aversion. Section III shows how these financial mistakes can compound and lead to substantial losses. Section IV shows the negative consequences of debt-biased behavior for borrowing decisions. We then develop and estimate a stylized model of debt aversion in Section V and discuss possible extensions and the relation to present focus in Section VI. Finally, we summarize and discuss our findings.

I. Experimental Design

We aim to show that debt causes financial mistakes by randomly assigning debt to participants. Outside the lab, there are many different types of debt with varying financial consequences. In our setting, debt simply represents a negative balance. Even this simplified and stylized version of debt, with no consequences for risk or wealth, can still impact financial decisions. As with any lab experiment, control and simplification come at the expense of generalization. In contrast to the contexts in previous empirical work, our environment allows us to identify causal effects and rule out confounding factors such as income uncertainty or the selection of borrowers. While we are aware of the experimental tradeoff, showing the negative impacts of debt aversion in the lab and providing a tool to identify debt-averse subjects is a necessary step toward better understanding the impacts of debt on financial decisions.

A. Basic Setting

Subjects own virtual accounts with different interest rates and balances that generate returns over time. The balances change based on subjects' decisions, while the interest rates are constant throughout the experiment. Accounts with positive balances generate positive returns, and accounts with negative balances generate negative returns. For example, an account with a starting balance of 1,100 points and an interest rate of 20 percent generates a positive return of 220 points. Similarly, an account with an initial balance of -900 points and an interest rate of 10 percent generates a negative return of -90 points. We label accounts with positive starting balances savings accounts and those with negative starting balances debt accounts. In the main setting, subjects can impact balances only by allocating points, not by redistributing balances, but we relax this restriction in additional treatments. Points allocated to an account increase its balance and affect the subsequent returns that it generates.

Subjects make four decisions over a week-long period. The timeline works as follows. For their first decision, participants must allocate an initial endowment of points. Two days after the first allocation decision, returns materialize, and these returns constitute the new endowment that subjects must allocate across their accounts. These subsequent allocation decisions likewise affect the balances and

hence the returns that accounts generate. This process repeats for one week until subjects have made four allocation decisions in total.

For a profit-maximizing individual deciding how to allocate points, the only relevant factor is the interest rate that each account provides. In all treatments in our experiment, the interest rates are kept constant; hence, all treatments have the same investment opportunities through allocation of points. For example, all treatments have accounts with 20 percent and 5 percent interest rates. While allocating 500 points to the 20 percent interest rate accounts generates a return of 100 points, allocating those points to the 5 percent account would generate only 25 points. It does not matter what the starting or current balances are or whether these accounts have a savings or debt label; only the interest rates matter for maximizing returns.

Since final payments occur at the end of the experiment and returns are certain, the action that maximizes monetary payoffs is unique regardless of risk or time preferences.⁷ Unlike other empirical settings where risk or return uncertainty can impact debt decisions, our experiment rules out these confounding factors; subjects cannot default on their debt, and investment returns do not fluctuate. Accounts accrue interest over time, and the final payments depend on the outstanding account balances.⁸ Paying subjects at the end of the experiment allows us to rule out other potential confounding effects like self-control (Cadena and Keys 2013) or present focus (Meier and Sprenger 2010; Allcott et al. 2022).

B. *Decisions with Debt: Main Treatments*

We introduce debt to subjects by giving them accounts with initial negative balances. However, the highest interest rate (20 percent) is always associated with a savings account. Subjects own six accounts, four of which they can allocate points to and two that we lock throughout the experiment. Locked accounts still generate returns, but participants cannot allocate them any points. Varying the starting balances of these locked accounts allows us to introduce debt while keeping initial wealth and returns constant across treatments.

In *No-Debt* treatment, subjects can allocate points only to savings accounts, providing a baseline for return-maximizing behavior. Since the action that maximizes returns is on the boundary of the action set (i.e., allocating all points to the highest-interest-rate account) we do not expect all subjects to choose that action.^{9, 10} The locked accounts correspond to Debt 1 and Debt 2, both with an initial balance of zero points and generating no returns. This feature allows us to control for the possibility that mentioning the word “debt” could affect choices. Hence, accounts

⁷We still elicit, and control for, risk and time preferences in our analysis using additional questions, which we discuss further in online Appendix B.

⁸We incentivize the final allocation decision by paying subjects the returns that their final balances generate without waiting two days.

⁹Previous research has shown that when the payoff-maximizing action is on the boundary, it is selected less often, e.g., in charity donation games. For a detailed discussion, see Vesterlund (2015).

¹⁰Other potential confounds could be balance-matching heuristics, as in Gathergood et al. (2019), or difficulties in finding the best interest rates, as in Ozyilmaz and Zhang (2019).

TABLE 1—ACCOUNTS IN MAIN TREATMENTS

	Savings 1	Savings 2	Savings 3	Savings 4	Debt 1	Debt 2
<i>Panel A. No-Debt</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	1,100	700	900	1,500	0	0
	Savings 1	Savings 2	Debt 1	Debt 2	Savings 3	Savings 4
<i>Panel B. Low-Debt</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	1,100	700	−900	−1,500	1,800	3,000

Note: In both cases, the sum of the balances is 4,200 points and the returns of these six accounts sum up to 500 points, which is the initial endowment that subjects must allocate.

labeled debt are present in all the main treatments. Since Savings 1 has the highest interest rate, allocating all points to it maximizes returns and the final payoff.¹¹

We introduce accounts with initial negative balances that generate negative returns in the *Low-Debt* treatment. These two accounts, Debt 1 and Debt 2, are not locked, and subjects can allocate points to them; to keep the same number of investment opportunities, we now lock two savings accounts, Savings 3 and Savings 4, with similar interest rates. We redistribute the initial balances such that the sum of the balances of Savings 3 and Debt 1, both with the same interest rate, is equal across treatments, and we do the same for Savings 4 and Debt 2. This redistribution allows us to introduce negative balances while keeping the net sum of balances, the number of actions, and the available interest rates equal across our main treatments. Despite the presence of the negative balances, the return-maximizing strategy does not change.

The parameters are such that subjects can obtain enough points during the experiment to repay both outstanding debt balances entirely. All subjects can zero out (fully repay) Debt 1 by the end of the second day, while Debt 2 requires three days.¹² Although subjects have negative starting balances, their initial wealth is positive, and their earnings can only grow.

We interpret deviations from the payoff-maximizing allocation as financial mistakes. While some of these could be due to the inherent complexity of the setting, the *No-Debt* and *Low-Debt* treatments allow us to isolate those mistakes caused by holding debt. Because the treatments have the same number and types of accounts and the same available interest rates, subjects go through the same set of instructions and understanding checks. Therefore, we do not expect differential understanding or treatment complexity to drive any treatment differences.

¹¹ Subjects make four allocation decisions so the interest difference can compound up to four times. The return on the initial 500 points, when allocated entirely to a 20 percent interest account, including the returns it generates, provides a $1.20^4 \approx 2.07$ return, as compared to the $1.05^4 \approx 1.22$ return from a 5 percent interest rate account.

¹² Due to additional opportunities to obtain points, subjects may be able to repay them earlier.

TABLE 2—ACCOUNTS IN HIGH-DEBT TREATMENT

	Savings 1	Savings 2	Debt 1	Debt 2	Savings 3	Savings 4
Interest rate (%)	20	10	15	5	15	5
Balance	1,100	700	−2,900	−3,500	3,800	5,000

C. Increasing the Negative Balances: High-Debt Treatment

In this treatment, we increase the initial negative balances so that subjects cannot fully repay them. We hypothesize that debt repayment strategies are contingent on the feasibility of repaying debt entirely. On the one hand, more debt could exacerbate deviations from payoff-maximizing behavior by devoting even more points to debt repayment. On the other hand, it could lead to better financial decisions. If subjects repay their debt only when they can repay it fully, we might observe more subjects focusing on maximizing their returns instead—which again means allocating all points to the highest-interest-rate account, Savings 1. Table 2 shows the available accounts and balances in the *High-Debt* treatment.

We increase the balances of the debt accounts (and both locked savings accounts) by 2000 points to maintain the same initial wealth and returns. These higher balances imply that most subjects will not be able to fully repay one debt account, even if they allocate all their earnings throughout the experiment.¹³

D. Redistribution Treatments

We introduce the option to reallocate balances from Savings 1 and Savings 2, giving subjects more agency over their accounts. By varying the accounts associated with low-interest rates, we can test whether redistribution decisions are also debt-biased.

In *Redistribution Debt*, participants can withdraw points from Savings 1 (20 percent) and Savings 2 (10 percent) accounts to increase their endowment of points. Account types are the same as in the *Low-Debt* but with different starting balances. While potentially beneficial, redistributing points could be detrimental to the subjects' returns. If they consolidate points in Savings 1, their returns will increase from 10 percent to 20 percent. However, they could instead use those balances to repay their outstanding debt. For example, moving points from Savings 1 to Debt 2 lowers the debt balance at the expense of reducing the return to 5 percent rather than the original 20 percent. We choose parameters such that if subjects reallocate all points towards Debt 2 at the beginning, the subsequent returns become so low that participants cannot fully repay any debt balance. If subjects maximize returns instead, they will accumulate enough points by day four to fully repay all debt at once if they want to.¹⁴ Hence, subjects have an additional option that is potentially beneficial but will severely decrease their final payoffs if misused.

¹³ Only subjects that manage to obtain substantial extra points from the additional questions can manage to repay one debt account entirely.

¹⁴ No single subject follows this strategy, although many participants attempt to repay their debt.

TABLE 3—ACCOUNTS IN REDISTRIBUTION TREATMENTS

	Savings 1	Savings 2	Debt 1	Debt 2	Savings 3	Savings 4
<i>Panel A. Redistribution Debt</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	2,000 (0)	400 (0)	−600	−4,300	300	6,400
	Savings 1	Savings 2	Savings 3	Savings 4	Debt 1	Debt 2
<i>Panel B. Redistribution No-Debt</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	2,000 (0)	400 (0)	600	4,300	−900	−2,200

Note: Minimum balance requirement in parentheses.

We control for the additional option to redistribute balances by running a *Redistribution No-Debt* treatment, where all accounts start with positive balances. Thus, redistribution cannot offset an outstanding negative balance. Table 3 shows that the available interest rates and initial net balances are comparable across treatments. We compensate for the positive balances by having locked accounts with negative balances. In both treatments, four accounts are savings, and two are debt accounts.

E. Borrowing Treatments

Finally, we modify our basic setting to allow subjects to increase their endowments by borrowing from their locked accounts. Hence, allowing us to see whether debt aversion also impacts borrowing decisions.

Participants in *Borrow-Debt* can borrow from Debt 1 and Debt 2 accounts that start with a zero balance. Thus, borrowing any amount causes balances to go negative. As a control, we run a *Borrow-Savings* treatment where subjects can borrow from accounts with positive starting balances. These starting balances are the same as the borrowing caps in the *Borrow-Debt* case, and subjects can borrow only until these accounts are zeroed out, thus resulting in the same borrowing cap as in the prior treatment. Table 4 shows how treatments only differ in the labeling and starting balances of some accounts, which do not affect the payoff-maximizing strategy.

In both borrowing treatments, the number of points that subjects may borrow is equal, with the only difference being the initial balance of the accounts and thus their labeling: savings or debt. Subjects make their borrowing decision each day before the allocation decision, and they can continue to borrow as long as the cumulative amount borrowed does not exceed the account caps, which are constant and equal across treatments. While subjects can borrow from these accounts freely up to the cap, the accounts are otherwise locked, and participants cannot allocate any points to them; i.e., they cannot repay any outstanding debt.

Because subjects can borrow from lower-interest-rate accounts, they can profit by properly investing these points, regardless of which borrowing treatment they belong to. However, this profitable opportunity is not mechanical, as it is possible for borrowing to reduce payoffs if subjects invest in even lower-interest-rate accounts. As in our main treatments, we maintain the same initial wealth, returns, and interest rates. One concern is that we are adding an action that subjects may or may not take

TABLE 4—ACCOUNTS IN BORROWING TREATMENTS

	Savings 1	Savings 2	Savings 3	Savings 4	Debt 1	Debt 2
<i>Panel A. Borrow-Debt</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	1,100	700	900	1,500	0 (−900)	0 (−1,500)
	Savings 1	Savings 2	Savings 3	Savings 4	Savings 5	Savings 6
<i>Panel B. Borrow-Savings</i>						
Interest rate (%)	20	10	15	5	15	5
Balance	1,100	700	0	0	900 (0)	1,500 (0)

Note: Minimum balance (i.e., borrowing caps) requirement in parentheses.

for other, nondebt reasons such as complexity. Hence, we compare subjects only across the two borrowing treatments and not against our main treatments.

F. Procedures

Subjects are recruited on Amazon Mechanical Turk (MTurk) and asked to complete an online survey for a week-long study. All participants are required to reside in the United States, to have completed at least 50 HITs with a 90 percent approval rate, and to not have participated in a past or similar version of this experiment. To prevent attrition, reminders are sent every six hours on the days of the allocation decisions. We recruit a total of 578 participants with unique IP addresses, and a completion rate of 85 percent, with a nonsignificant difference in attrition by treatment.¹⁵ Treatment assignment happens only after the participant responds to the follow-up email from the initial survey, which ensures that participants receive the notifications and minimizes differential attrition by treatment.

Since all the treatments involve allocating points, instructions are identical in all of them. During the instruction period, participants view several examples and comprehension check questions that they must correctly answer before proceeding to the main decisions.¹⁶ We track the number of errors that subjects make and use this as a control in our analysis. Each decision day, subjects must go through the instructions again, including the comprehension check questions. Thus, by the last day, subjects have gone through the instructions four times altogether.

Subjects who finish the entire experiment are paid a \$10 participation fee and a bonus based on their performance. Payments are determined as follows: after the last allocation decision, all point balances are added up and converted into dollars with a 500-point-to-\$1 conversion rate. To incentivize the last allocation, we calculate the corresponding returns for the final balances and add them to the total point count in addition to payments from the risk and time elicitations. The average subject made \$34 and took two hours to complete all parts of the survey.

¹⁵ On average, each treatment has 82 participants, ranging from 77 to 86.

¹⁶ After the first allocation decision, we ask if any part of the instructions or the survey were confusing. A total of 94 percent of subjects mention no problem in understanding all parts.

At the conclusion of the experiment, we elicit a series of demographic and feedback questions.¹⁷ We obtain information on subjects' general characteristics such as age or gender and on their financial situation such as student loan exposure or outstanding debt. Beyond eliciting demographics, we ask subjects to describe the reasoning behind their decisions in their own words. Subjects describe their own strategy and how they would behave in hindsight given their knowledge of how the experiment has played out. In addition subjects evaluate, through a ranking from one to five, the relevance of different aspects (balances, interest rates, and debt) of their decision-making process. While these questions are not incentivized, they still provide suggestive evidence on participants' behavior and the underlying mechanism.

One-Shot Allocations.—In the last allocation decision, right before finishing the experiment, subjects are presented with three simplified one-shot scenarios mimicking the main treatments to assess the robustness of our week-long findings. Each one-shot scenario corresponds to one of the three main treatments, *No-Debt*, *Low-Debt*, or *High-Debt*, but with only one allocation decision instead of four. Subjects face these three decisions in random order, knowing that only one will count for payment. In all of these scenarios, subjects must allocate 1,000 points among the four available accounts.¹⁸ As in the week-long experiment, the payoff-maximizing action is to allocate all points to the account with the highest interest rate (20 percent).

II. Consequences of Holding Debt

Due to the similarities between our main treatments, we interpret differences in behavior as evidence of financial mistakes caused by debt. When considering participants who maximize returns in all four allocation decisions, we find almost three times more subjects in *No-Debt* than in *Low-Debt*. Subjects in *Low-Debt* focus instead on repaying their outstanding balances, with over one-third zeroing out at least one account. Using our *High-Debt* treatment, we show that when this strategy is no longer feasible, the effects are heterogeneous: more subjects maximize returns, while those who pay off debt end up repaying larger amounts.

A. The Opportunity Cost of Repaying Debt

Subjects with debt accounts are more likely to deviate from the return-maximizing strategy. As a first step, consider the allocation decision on day 1, where everyone has the same starting wealth, points to allocate, and financial incentives. Figure 1 shows that subjects in both treatments allocate the largest share of points to the account with the highest interest rate (Savings 1, 20 percent). However, while *No-Debt* subjects allocate 73 percent of their initial 500 points to Savings 1, *Low-Debt* subjects allocate less than half (47 percent) to that same account, a 26 percentage point difference (p -value < 0.001). This difference is driven mainly by *Low-Debt* subjects allocating 18 percentage points more to the debt account with the highest interest

¹⁷ See online Appendix A for details and a timeline of the design in online Appendix Table A.9.

¹⁸ See online Appendix Table A.11 for details on each one-shot scenario.

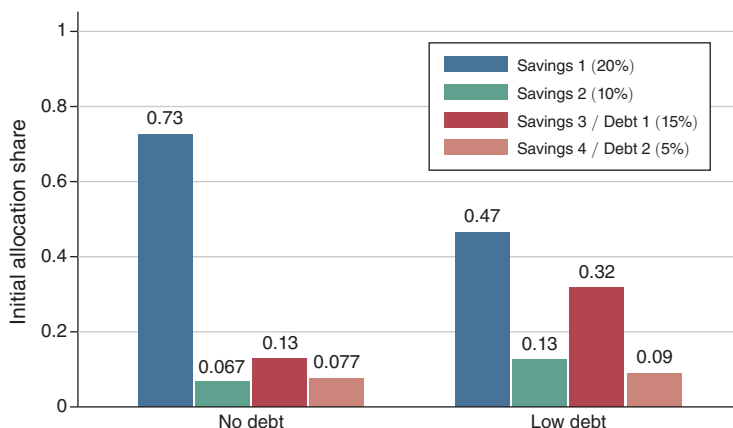


FIGURE 1. ALLOCATION SHARES OF THE INITIAL ENDOWMENT IN DAY 1

rate (Debt 1, 15 percent) than do *No-Debt* subjects to the account with the same interest rate (Savings 3, 15 percent, p -value < 0.001).¹⁹ Thus, we find that subjects with debt are less likely to allocate points to the accounts that would benefit them the most financially. These differences in behavior translate to lower returns; in *Low-Debt*, returns from the initial allocation are 9.9 percent lower. On average, *No-Debt* subjects obtain a 17.5 percent average return on their first allocation decision, as compared to 15.8 percent among *Low-Debt* subjects (p -value = 0.001).

In an additional robustness treatment, we show that the reason that participants focus on repaying Debt 1 is that it has the higher interest rate and not because it has a lower outstanding balance. Arguably, participants could have chosen that account because the lower balance makes it easier to pay it faster. We rule out this alternative explanation by running a robustness treatment with 80 additional subjects in which we swap the balances of the debt accounts while keeping the interest rates constant. Now, the 15 percent interest account also has a larger balance. This exercise does not alter our results: we still find that 28 percent of the initial endowment is allocated to repaying Debt 1, similar to our previous finding of 32 percent.²⁰

Taking into consideration all four days of our experiment, we still find more deviations from the return-maximizing strategies among subjects who hold debt. Thirty-eight percent of subjects in *No-Debt* maximize returns in all decisions, while the rate drops down to 13 percent in *Low-Debt*, a 25 percentage point difference (p -value < 0.001).

We interpret these results as evidence that subjects do not behave randomly but instead make debt-biased financial mistakes. Despite a lower share of subjects maximizing returns in the *Low-Debt* case, most subjects allocate points to the account with the second-highest interest rate. In fact, the *Low-Debt* ranking of the allocation shares directly follows the ranking of the interest rates. Furthermore, we find limited evidence that subjects use other strategies present in the literature, such as

¹⁹There is also an increase in the allocation to the other savings account (Savings 2, 10 percent), but of smaller magnitude (6 percentage points, p -value = 0.003).

²⁰We provide further details on this robustness treatment in online Appendix A.9.

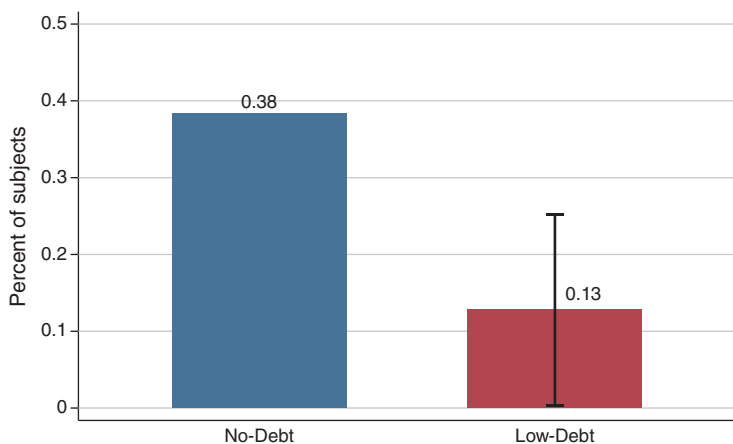


FIGURE 2. PERCENT OF SUBJECTS THAT MAXIMIZE RETURNS IN ALL DECISIONS

Note: Error bar corresponds to the 95 percent confidence interval on the difference across treatments.

equal-split or balance-matching heuristics.²¹ Since Debt 1 has a lower balance in absolute value than Debt 2, balance-matching strategies cannot explain why subjects predominantly allocate points to Debt 1. In the next subsection, we further explore what strategies subjects follow when holding debt.

B. Repayment Strategies When Subjects Hold Debt

Given the evidence on how subjects with debt deviate more often from return-maximizing behavior, we now explore the more prevalent strategies subjects employ when holding debt. One hypothesis that we aim to rule out is that subjects with debt behave more erratically, i.e., that they allocate points randomly without considering interest rates. Another possibility is that subjects focus first on repaying their debt and then switch to maximizing returns.

We find evidence that subjects' strategies are contingent on outstanding negative balances. Figure 3 depicts the total number of points allocated to debt accounts in the *Low-Debt* case and their equivalent savings accounts in the *No-Debt* case. While in the latter, 38 percent of subjects allocate no points to the account with the second-highest interest rate, only 13 percent of the *Low-Debt* subjects follow the same strategy (p -value < 0.001). We observe a similar pattern for the lowest-interest-rate account, with 51 percent of subjects allocating no points in the *No-Debt* case versus 28 percent in the *Low-Debt* case (p -value = 0.002). Furthermore, we find the largest difference when focusing on subjects who allocate a nonzero amount to the debt accounts and their equivalents in the *No-Debt* treatment. Large fractions of subjects, 34 and 17 percent, exactly zero out the outstanding negative balances of Debt 1 and 2, respectively. These spikes are not mechanical, as subjects can still allocate

²¹In Gathergood et al. (2019), debt repayments are consistent with a balance-matching heuristic under which the share of repayments is matched to the share of the balance.

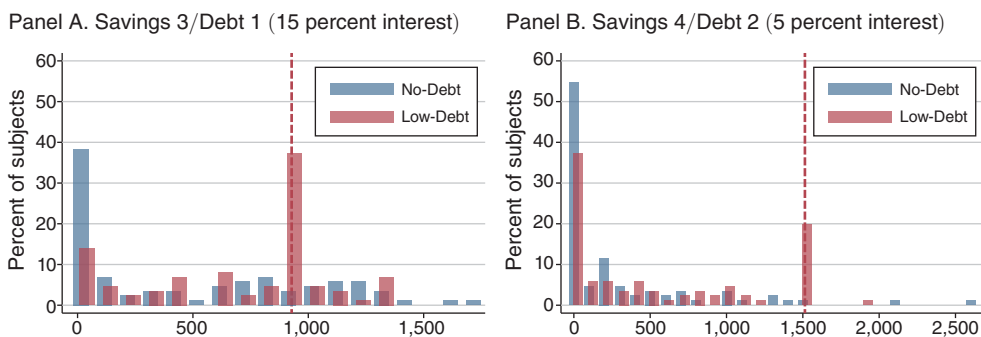


FIGURE 3. TOTAL POINTS ALLOCATED TO 15 PERCENT AND 5 PERCENT ACCOUNTS

Note: Vertical dashed lines indicate allocations that offset the negative balance for debt accounts.

points after fully repaying the debt balances.²² We do not observe the same pattern for the equivalent accounts in the *No-Debt* case.

This evidence indicates that many participants perceive negative balances differently and that this perception, rather than erratic behavior, causes debt-specific strategies. More than one-half of the *Low-Debt* participants (54 percent) fully repay the outstanding negative balance associated with the higher interest rate, with 34 percent zeroing it out exactly. Among those participants who repay the low-interest-rate debt, representing 21 percent of the sample, almost all (94 percent) also repay the high-interest-rate debt. These results provide compelling evidence that participants use somehow sophisticated strategies when focusing on repaying their debts. They give higher priority to high-interest-accruing debt accounts. Once that balance is offset, some subjects proceed to pay their remaining low-interest debt, while others switch to a strategy of maximizing returns.

We provide additional evidence of repayment patterns using subjects' strategy descriptions on the last day. In their descriptions, 29 percent of the *Low-Debt* subjects explicitly mention using a strategy of first repaying their outstanding debt and then focusing on maximizing returns. These hindsight questions are not incentivized; however, most descriptions are consistent with the observed behavior in the experiment. See online Appendix E for example answers from subjects.

In the next subsection, we use our *High-Debt* treatment to explore what strategies subjects follow when negative balances cannot be fully repaid. Do subjects make even larger financial mistakes or do they ignore debt accounts and focus on maximizing returns?

C. The Effects of Larger Debt Balances: High-Debt Treatment

We observe heterogeneous effects in behavior when negative balances cannot be fully repaid: more subjects maximize returns in all decisions, but those who focus on repaying debt end up allocating even more points to debt than in *Low-Debt*.

²² Figure 3, panel A shows that a small number of subjects end up with debt accounts with positive balances. In those cases, the accounts go from generating negative to positive returns.

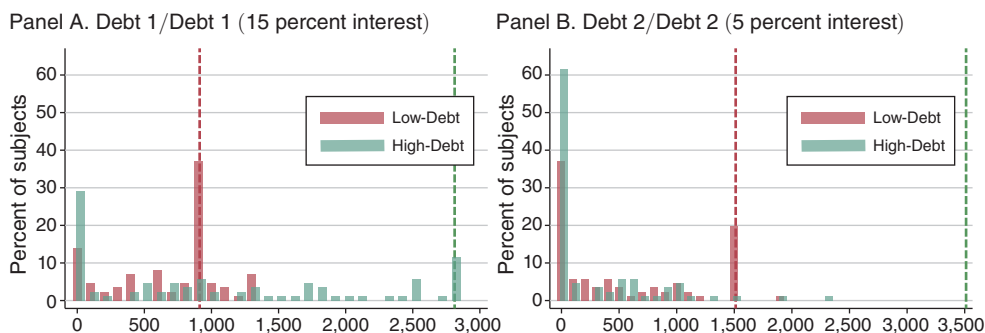


FIGURE 4. TOTAL POINTS ALLOCATED TO DEBT ACCOUNTS

Note: Vertical dashed lines indicate allocations that offset the negative balance for debt accounts.

Now, 26 percent of *High-Debt* subjects maximize returns in all decisions—a larger proportion than the 13 percent in *Low-Debt* (p -value = 0.045) but still lower than the 38 percent in *No-Debt* (p -value = 0.052). Compared to subjects in *Low-Debt*, we find fewer subjects allocating points to either debt account. Figure 4 shows that when initial negative balances are larger, 27 percent of subjects do not allocate any points to Debt 1 and more than 56 percent do not allocate any points to Debt 2.

The fact that we observe lower deviations from return-maximizing behavior in *High-Debt* could lead us to conclude that larger debt balances reduce the financial impacts of holding debt. However, we find that the average measure masks heterogeneity in our results. Despite the increase in return-maximizing behavior, some subjects allocate more points to debt repayment. *High-Debt* subjects who do not maximize returns allocate a larger share of their total points to debt repayment than *Low-Debt* subjects who do not maximize returns (8.9 percentage points, p -value = 0.017). Furthermore, 10 percent of participants fully repay the debt account with the highest interest rate (Debt 1), which has an initial outstanding balance of 2,900 points. This implies that these subjects allocate most, if not all, of their returns to this account.²³

Our evidence shows that subjects' strategies depend on the size of the initial debt balances. When debt balances cannot be fully repaid, more subjects maximize returns, but those who do not perform even worse. If subjects consider only interest rates, behavior in the *Low-Debt* and *High-Debt* treatments should not differ, as the only difference is the starting balances, which do not affect the payoff-maximizing strategies. This evidence suggests a sort of debt-free premium, implying that balances should be considered in modeling behavioral responses to debt.²⁴ Furthermore, this could indicate that some subjects might be partially sophisticated concerning

²³The fact that we do not observe a similar spike around 900 or 1,500 points as in the *Low-Debt* case also suggests a "discontinuity" when negative accounts reach a zero balance.

²⁴In Section V, we briefly introduce and discuss the implications of a model where subjects derive negative utility from holding debt balances.

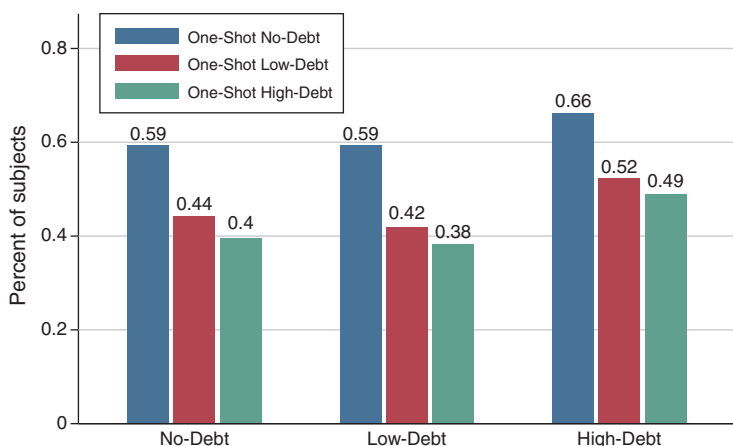


FIGURE 5. PERCENT OF SUBJECTS WHO MAXIMIZE RETURNS IN ONE-SHOT SCENARIOS

debt repayments and allocate points only if they can repay outstanding balances completely.²⁵

D. Robustness: One-Shot

We replicate our main result using the three one-shot scenarios that subjects face after the last allocation decision. Here, subjects face three one-shot versions of the main allocation decisions, all on the same day, in random order. We find that regardless of their previous experience and treatment assignment, subjects are more likely to maximize returns in *One-Shot No-Debt* than in any of the other one-shot scenarios with debt.

Among subjects that completed the *No-Debt* treatment, 59 percent maximize returns in the *One-Shot No-Debt* scenario. In contrast, the percentage of subjects facing the *One-Shot Low-Debt* scenario who maximize returns is 15 percentage points lower (p -value < 0.001). A similar difference arises in the *One-Shot High-Debt* scenario (p -value < 0.001).²⁶

When we look at the subjects who completed either of the other treatments, *Low-Debt* or *High-Debt*, a similar pattern arises. More subjects maximize returns in the *One-shot No-Debt* scenario than in the *One-Shot Low-Debt* and *One-Shot High-Debt* scenarios. Despite previous experience with debt accounts, these subjects behave similarly, with no statistically significant differences based on initial treatment assignment.

²⁵This is consistent with previous experiments showing that subjects exhibit a premium for closing negative balances; see Amar et al. (2011); Besharat, Carrilat, and Ladik (2014); Kettle et al. (2016). In addition, Zhang et al. (2020) show how people disengage from debt when full repayment seems difficult.

²⁶Note that unlike our in main treatments, we do not observe a significant difference between the *One-Shot Low-Debt* and *One-shot High-Debt* groups.

TABLE 5—REDISTRIBUTION DECISIONS: SHARE OF SUBJECTS WHO CONSOLIDATE ON DAY 1

	Redistribution No-Debt	Redistribution Debt	<i>p</i> -value
Savings 1	37.04	16.88	0.004
Savings 2	3.7	2.60	0.692
Savings 3/Debt 1	13.58	33.77	0.003
Savings 4/Debt 2	8.64	12.99	0.384

Notes: Subjects are assigned to one category if they allocate more than the initial endowment to that account. This is only feasible if subjects redistribute balances from Savings 1 or Savings 2. With the exception of Savings 1, subjects can be classified in more than one category.

III. Increased Agency and Compounded Financial Losses

When subjects can reallocate their savings balances, debt-biased behavior in redistribution and allocation decisions compounds and exacerbates financial losses. One-third of subjects use redistribution to repay their debts quickly, and 20 percent of the participants end up with a lower return than the worst possible allocation without redistribution. Furthermore, we find that subjects who attempt to repay their debt do so suboptimally. Although feasible, no subject fully manages to repay their debt.

A. Redistribution Decisions

Subjects with debt are less likely to use redistribution decisions to maximize their returns and instead focus on repaying debt. Table 5 shows that despite similar uptake, there are significant differences in how subjects employ these redistribution options. Since subjects have an initial endowment of 500 points, any allocation above 500 on the first day is feasible only through redistributing balances. Hence, we classify each subject as consolidating toward an account if she allocates more than 500 points to it. In the *Redistribution No-Debt* group, individuals are twice as likely to redistribute balances towards the high-interest-rate account (37 percent versus 17 percent; p -value = 0.004), maximizing returns. On the other hand, 34 percent of subjects in *Redistribution Debt* consolidate towards Debt 1 (15 percent interest, –600 initial balance). For the other two accounts, Savings 2 and Savings 4/Debt 2, we find no significant differences.

These results suggest a sophisticated strategy contingent on having debt, in line with our previous results: around 20 percent of participants consolidate to maximize returns when debt-free but concentrate on repaying high-interest debt when assigned to *Redistribution Debt*. Furthermore, the lack of treatment differences in the other accounts suggests that confusion or lack of understanding is comparable across conditions and unlikely to explain our results.

In addition to differences in redistribution decisions, we replicate our allocation results from the main treatments: subjects holding debt accounts are less likely to allocate points toward the account with the highest interest rate. When we classify subjects based on their allocation decisions, similarly to what we identify in the main treatments, we find that 43 percent of subjects exclusively allocate points to

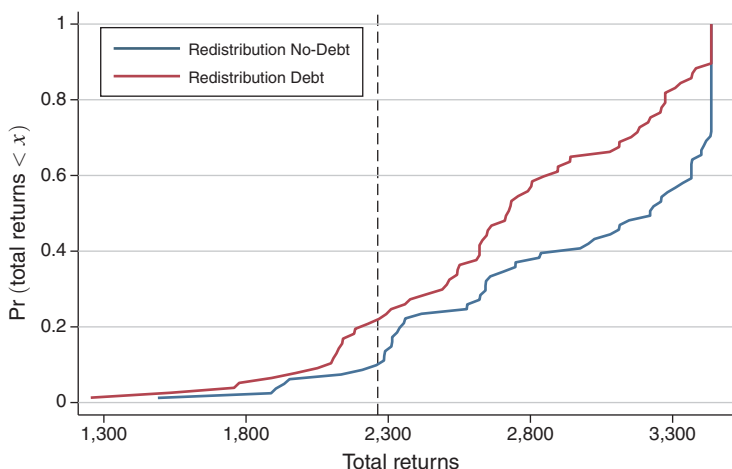


FIGURE 6. TOTAL RETURNS

Note: Vertical dashed line indicates the return from the worst allocation without redistribution.

the highest-interest-rate account in the *Redistribution No-Debt* case but only 14 percent in *Redistribution Debt* case in all decisions (p -value < 0.001).

We find similar redistribution behavior once we control for the optimality of allocation decisions. In both treatments, approximately three-quarters of returns maximizers also redistribute as much as possible towards Savings 1 (66 percent versus 82 percent, p -value $= 0.198$), maximizing final payoffs. This pattern is perhaps not surprising, as redistribution involves only accounts with positive balances in both treatments.²⁷

B. Compounding Debt-Biased Behavior

Debt-biased redistribution and allocation decisions compound and lead to subjects obtaining lower returns in *Redistribution Debt*. Figure 6 depicts two main differences when we focus on total returns throughout the experiment. First, subjects in *Redistribution Debt* are less than half as likely to obtain the maximum attainable return (30 percent versus 12 percent, p -value $= 0.005$).²⁸ Second, subjects with debt are more likely to obtain lower returns than they would have obtained under the worst possible allocation. Nonetheless, this difference is only marginally significant (21 percent versus 10 percent, p -value $= 0.059$).²⁹ This last result implies that some participants would have benefited had we removed the option to redistribute balances.

²⁷ We study the implications of altering this feature in our two *Borrowing* treatments.

²⁸ This return corresponds to the strategy of consolidating towards Savings 1 on day 1 and allocating all the endowment and subsequent returns to Savings 1 as well, which results in 3,436 points.

²⁹ The total returns for the worst possible allocation decision corresponds to 2,263 points. This strategy allocates the initial endowment and subsequent returns to the account with the lowest interest rate (5 percent) without redistributing any points.

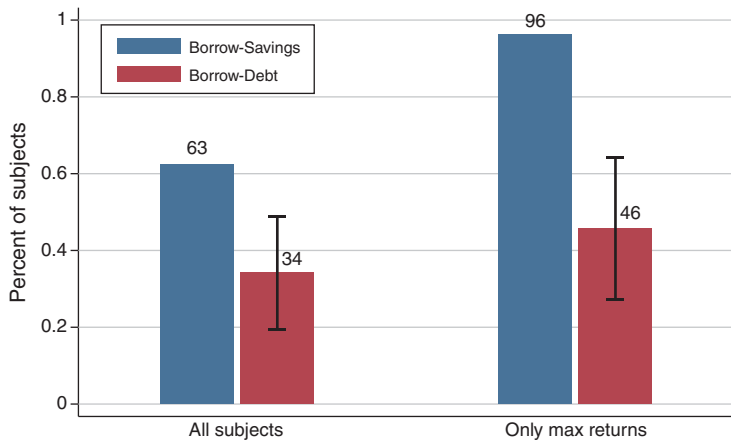


FIGURE 7. PERCENT OF SUBJECTS WHO BORROW THE MAXIMUM AMOUNT FROM BOTH ACCOUNTS

Notes: Only Max Returns restricts to subjects who allocate all points to the highest interest account in all decisions. Error bars correspond to the 95 percent confidence interval on the difference across treatments.

Although it is feasible, we find that no subject fully repays her outstanding debt balances. For our given parameters, a subject who consolidates into Savings 1 and allocates all her points to it would start day four with an endowment of 922 points and a balance of 4,308 points in Savings 1. These amounts are enough to completely repay all outstanding debt (4,900 points) before the end of the experiment. However, we find that no subjects follow such a strategy. Instead, we find that participants focus on repaying their debt accounts quickly at the cost of lower returns and ending the experiment with outstanding debt balances.

IV. Reluctance to Borrow

When allowed to borrow, participants can make two financial mistakes: first, they can be reluctant to borrow and miss a profitable opportunity, and second, they can borrow but misallocate the funds. We find evidence only in support of the former. Subjects are reluctant to borrow from debt accounts, consistent with debt aversion, but those who take advantage of the borrowing opportunities are not negatively selected and obtain high returns. Since borrowing opportunities, regardless of treatment, can always provide a gain by moving points from lower interest accounts (15 percent and 5 percent) to a 20 percent interest rate account, subjects who choose not to borrow end up missing out on these potential gains.

A. Borrowing Decisions

The holding of debt severely impacts borrowing decisions: 34 percent of subjects in *Borrow-Debt* borrow the maximum amount, compared to 63 percent in *Borrow-Savings* (a 29 percentage point difference, p -value < 0.001). That is, changing the label of the account from which participants borrow from savings to debt reduces the likelihood of borrowing the maximum amount by almost 50 percent. These results are unlikely to be driven by confusion or a lack of understanding that

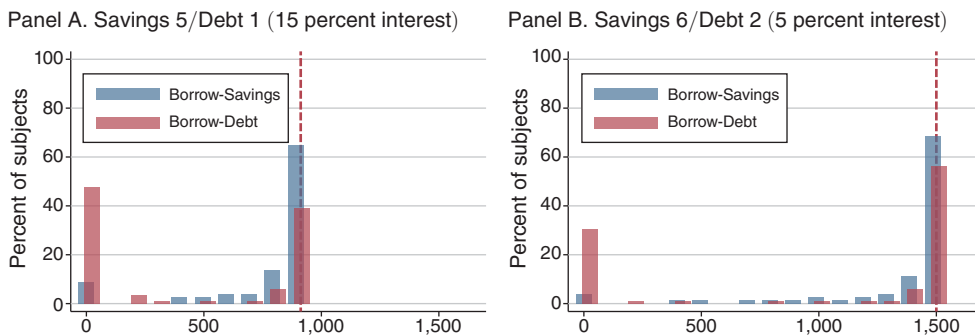


FIGURE 8. TOTAL POINTS BORROWED FROM EACH ACCOUNT

Note: Vertical dashed lines indicate the borrowing limits for each account.

TABLE 6—BORROWING TREATMENTS: RETURN AND PAYMENT ESTIMATION OUTPUT

Sample:	Log total returns		
	All subjects (1)	Max returns (2)	All subjects (3)
Mean of dep.var.	8.618 (0.078)	8.681 (0.085)	8.498 (0.051)
Borrow-Debt	−0.053 (0.027)	−0.086 (0.033)	0.004 (0.022)
Borrow Max			0.182 (0.022)
Observations	162	61	162

Notes: Results from a linear regression with robust standard errors in parentheses. The dependent variable is the log of the cumulative returns in the four allocation decisions and the final payments that subjects obtained without the participation fee. Borrow Max is a dummy that equals 1 if the subject borrowed the maximum amount by the end of the experiment. The regression also includes several controls, including demographic characteristics. See online Appendix Table A.19 for the full output.

borrowing could be profitable: when we restrict the sample to subjects who maximize their returns in all allocation decisions, the difference in borrowing behavior persists. Figure 7 shows that within this restricted sample, 96 percent—almost the entirety of the subsample—borrow the maximum amount from both accounts in *Borrow-Savings*. In contrast, not even one-half of the subjects in *Borrow-Debt* (46 percent), who otherwise maximize returns, borrow the maximum amount from both accounts. Thus, we see that debt-biased behavior still causes financial mistakes in borrowing decisions, even for those aware of the high returns that they could obtain.

We observe a bimodal distribution in the amounts that participants borrow from each account. Figure 8 shows how most subjects borrow either the maximum amount or nothing at all. In *Borrow-Savings*, 64 percent of subjects borrow up to the cap from the high-interest-rate account, and 66 percent do the same from the low-interest-rate account. In contrast, subjects in the *Borrow-Debt* group are almost half as likely (38 percent) to borrow the maximum from their 15 percent account, and 52 percent hit the borrowing cap from the 5 percent account. We can also see these treatment

differences by looking at how many participants borrow nothing. Only 8 percent of subjects in the *Borrow-Savings* group borrow nothing from the high-interest-rate account, but 46 percent of subjects borrow nothing from the equivalent account in *Borrow-Debt*. This same qualitative difference persists for the low-interest-rate account, albeit not as stark.

Consistent with results from Section II, *Borrow-Debt* participants seem to consider interest rates when deciding which account they borrow from. While in *Borrow-Savings* the number of participants who borrow the maximum amount for each account is almost equal (64 versus 66 percent, p -value = 0.3204), we find a significant difference for the two accounts (38 versus 52 percent, p -value = 0.004) in *Borrow-Debt*. It is worth mentioning that unlike in our main treatments in Section II, where the debt accounts differ in their initial balances, in this case, the two accounts have the same initial balances. The only differences between the two are the interest rates and the borrowing caps. Overall, this evidence suggests heterogeneity in the degree of debt aversion that subjects exhibit. Approximately 45 percent of the subjects are unwilling to borrow from a high-interest-rate account, but half of them borrow when the interest rate is low.

Given the disparity in borrowing behavior, we also find differences in the average returns that subjects accrue. By the end of the study, *Borrow-Savings* subjects have 5 percent higher returns than *Borrow-Debt* subjects (p -value = 0.047). These differences are explained entirely by different borrowing behavior and not by different allocation decisions. Once we control for subjects borrowing the maximum amount, there is no longer a difference in returns across treatments (p -value = 0.854). This is expected since all accounts subjects can allocate points to have an initial positive balance.

One possible explanation for the reluctance to borrow is that sophisticated subjects anticipate that after they borrow and hence have debt, they will make worse allocation decisions similar to those that we find in our previous treatments. However, we rule out this possibility by design, as subjects cannot allocate any points to the accounts that they borrow from. Even in the *Borrow-Debt* treatment, subjects are allowed to allocate points only to savings accounts. Thus, concerns about anticipated debt-biased mistakes cannot explain the reluctance to borrow that we observe.

Taking our findings together, we find evidence of debt-averse behavior that prevents subjects from taking advantage of a profitable opportunity. These financial mistakes apply to many subjects, even those who otherwise maximize returns in their allocation decisions. Whereas in our main treatments, debt-biased behavior manifests as financial mistakes in debt repayments, we now show that financial mistakes are also present in borrowing decisions, in line with previous findings in the debt-aversion literature.³⁰

B. Selection of Borrowers

Although we provide borrowing opportunities that are easy to profit from, it is still possible for borrowers to make financial mistakes by misallocating the additional

³⁰ For example, there is reluctance to borrow among prospective college students (Callender and Jackson 2005), low-wage workers (Gopalan et al. forthcoming), and bank customers Caetano, Palacios, and Patrinos. (2011).

funds. We find this is not the case: borrowers are positively selected. Those who borrow the maximum amount from both savings accounts allocate 87 percent of their total points to the highest-interest-rate account, and those who borrow from debt accounts allocate 79 percent. This result holds when we look at how many participants maximize returns in all allocation decisions: 50 and 57 percent of subjects who borrow the maximum also maximize returns in all decisions in *Borrow-Savings* and *Borrow-Debt*, respectively, a nonsignificant difference ($p\text{-value} = 0.106$).

While our initial treatment groups relied on endowing subjects with debt to find treatment effects, we have again found evidence of debt-biased behavior when indebtedness is chosen rather than inherent. In this borrowing scenario, subjects do not start with debt. Nevertheless, we still find that debt has an impact through the reluctance to borrow from debt accounts, ultimately leading to lower final payoffs.

V. Quantifying Debt Aversion

In this section, we develop and estimate a stylized model in which agents overweight debt. At the individual level, we find substantial heterogeneity in debt attitudes. Among our main treatments sample, one-third of our subjects exhibit behavior not indicative of debt aversion, while a quarter behave in a manner consistent with some degree of debt aversion. This latter subsample encompasses those with low debt aversion, who overweight debt by at least 4 percent, and those with high debt aversion, who overweight debt by at least 14 percent. On average, for subjects who we can classify according to our model with a small margin of error, we find that they will trade off a dollar less of debt for 1.03–1.07 dollars in savings.

A. A Simple Model of Debt Aversion

We envision a consumer who makes several allocation decisions over T periods. In each period t , her task is to allocate a given endowment X^t across a finite set of accounts: $J = 1, \dots, n$. Each of these accounts has a starting balance b_j^t and generates a return of $b_j^t i_j$ in the next period, where the interest i_j is time invariant and positive. In a similar fashion as in our experimental design, endowments are determined by the sum of the accounts' returns realized in that period. As time passes, balances reflect previous allocations so that these are intertwined. Formally, the agent evaluates a particular allocation $\sigma = ((x_1^1, \dots, x_J^1), \dots, (x_1^T, \dots, x_J^T))$ as

$$U(\sigma) = \sum_{t=1}^T \sum_{j=1}^J (b_j^t + x_j^t)(1 + i_j),$$

where x_j^t is the amount allocated to account j in period t . We further impose that allocations must exhaust the endowment $\sum_{j=1}^J x_j^t = X^t, \forall t \in T$ and that agents cannot redistribute balances $x_j^t \geq 0, \forall t \in T, \forall j \in J$.

We introduce debt aversion by making subjects overweight accounts with negative balances. In particular, we assume that agents have a debt aversion parameter λ that

distorts negative balances, i.e., $b_j^t + x_j^t < 0$.³¹ Under this additional assumption, the agent now evaluates allocations as follows:

$$(1) \quad U(\sigma) = \sum_{t=1}^T \sum_{j=1}^J (b_j^t + x_j^t)(1 + i_j)(1 + \lambda \mathbf{1}\{b_j^t + x_j^t < 0\}).$$

Taking the first-order conditions, we find the following expressions:

$$\begin{aligned} \frac{\partial U}{\partial x_j^t} &= 1 + i_j \quad \text{for } b_j^t + x_j^t \geq 0, \\ \frac{\partial U}{\partial x_j^t} &= (1 + i_j)(1 + \lambda) \quad \text{for } b_j^t + x_j^t < 0. \end{aligned}$$

Conveniently, these expressions do not contain any term that depends on subsequent decisions, and hence, this maximization problem can be solved separately for each period. Therefore, we can drop any time indexes and focus directly on the one-period problem. This simplification is a direct consequence of the specific functional form that we assume for debt aversion. When discussing our results, we will also consider an alternative model that adds a fixed cost for indebtedness to match our findings in the high-debt and borrowing treatments.³² For exposition purposes, let us assume that only four accounts are available: two of them with positive initial balances ($b_1 \geq 0$ and $b_2 \geq 0$) and the two others with negative balances ($b_3 < 0$ and $b_4 < 0$). In addition, let the highest interest rate correspond to one of the positive balance accounts, for example, i_1 .

Despite its simplicity, the model generates rich dynamics. Agents with low enough values of λ simply maximize their returns by allocating everything to the highest-interest-rate account in every period.³³ For a sufficiently high λ , subjects instead allocate their full endowment to the negative balance account with the highest interest. These agents keep doing so until they eventually repay the negative balance, and switch to maximizing total returns. We refer to these as *Low* λ agents. Finally, for very high values of λ , agents first attempt to repay all of their negative balances accounts, as opposed to just the highest-interest one, before switching to maximizing returns. We denominate this last group *High* λ .

Applying the previous logic to our experimental design yields three λ ranges, each one corresponding to a different strategy. The first one is for subjects with $\lambda < 0.05/(1 + 0.15) \approx 0.04$. These are subjects who either do not over weight debt at all or do so little enough that it does not impact their choices. For λ values above $0.05/(1 + 0.15)$ but smaller than $0.15/(1 + 0.05) \approx 0.14$, subjects repay only one debt account, and for $\lambda > 0.15/(1 + 0.05)$, subjects first repay both of their debt accounts before focusing on the highest-interest-rate account.

³¹Consistent with our experimental evidence, we assume that this debt overweighting applies to the current balance even if repayments are not due until later periods.

³²For example, as we show in the *High Debt* treatment, the feasibility of zeroing out a debt balance seems to matter for subjects' decisions to start paying their debts in the first place. A different version of our model, consistent with this evidence, could incorporate a debt-free premium, calibrated so a fraction of participants find repaying their debt worthwhile only if they are able to repay it fully. In that case, current allocation decisions would consider future choices too. For simplicity of exposition, we discuss this possibility later on in the results and its implications.

³³Specifically, $\lambda < (i_1 - i_{j'})/(1 + i_{j'})$, where $i_{j'}$ corresponds to the highest interest rate of a debt account.

TABLE 7—ESTIMATES OF REPRESENTATIVE LAMBDA

	Average $\hat{\lambda}$	Average $\hat{\lambda}$ (noise adjusted)
<i>Panel A. Main treatments</i>		
No-Debt	0.021	0.000
Low-Debt	0.043	0.034
<i>Panel B. Borrowing treatments</i>		
Borrow-Control	0.047	0.009
Borrow-Debt	0.075	0.067
<i>Panel C. Redistribution treatments</i>		
Redistribution Control	0.033	0.001
Redistribution Debt	0.048	0.025
<i>Panel D. Other treatments</i>		
High-Debt	0.014	0.014
One-Shot	0.029	0.026

Notes: Noise adjusted corresponds to the average $\hat{\lambda}$ restricted to participants with an MSE no larger than 0.054. This refers to the MSE threshold for which no subject is mistakenly classified as debt-averse in the *No-Debt* treatment. By construction, the noise-adjusted average $\hat{\lambda}$ for *No-Debt* is 0. We report significance tests using a bootstrap method in online Appendix I.

B. Estimation

Using the different ranges of λ , we can predict how agents of each λ type allocate their endowment based on the given balances. We compare these predictions to the actual responses and see which type best predicts choices, i.e., we perform minimum distance estimation. We apply the same procedure for all four allocation decisions that subjects make and calculate the mean squared error (MSE) of each type's prediction. To ensure that our measure of fitness does not depend on the size of the endowment, which varies across decisions, we calculate deviations using endowment shares.

Table 7 reports the average $\hat{\lambda}$ for each treatment. Subjects in *Low-Debt* exhibit large heterogeneity in their debt-aversion attitudes.³⁴ 50 percent show no signs of debt-averse behavior, while the remaining half is evenly split among low and high levels of over weighting. This translates into an average $\hat{\lambda}$ of 0.043. As a placebo, we repeat the same classification exercise but using allocations from the *No-Debt* treatment. In other words, how allocations in *No-Debt* would be classified had they occurred in *Low-Debt*.³⁵ This exercise reveals that while there is a substantial fraction of subjects whose behavior is not well explained by either type,³⁶ there are few instances where subjects are accidentally classified as debt averse. Specifically, for an MSE of 0.054, no subject in *No-Debt* is mistakenly classified as debt averse. Therefore, we also report in Table 7 estimates at that MSE threshold for all treatments. Using this new threshold, 48 percent of our subjects are classified in

³⁴ See online Appendix I for details on the individual distribution of $\hat{\lambda}$ types.

³⁵ To make this exercise comparable to our previous one, we decrease the balances of the Savings 3 and Savings 4 accounts by 1,800 and 3,000 points, respectively, and run the same code as in the *No-Debt* treatment. See online Appendix Figures A.11 and A.12 for the distributions of types.

³⁶ A fraction comparable to *Low-Debt*, suggesting a similar presence of “noisy participants.”

Low-Debt.³⁷ Nonetheless, the relative proportions of classified types are fairly similar: 58 percent do not overweight debt at all, and 42 percent are classified as debt averse, averaging 0.035.

Classification of participants in our borrowing treatments reveals a similar pattern. A large fraction of participants is consistent with debt-averse behavior, with a noise adjusted $\hat{\lambda}$ of 0.067. While we find almost no debt-biased decisions in *Borrowing Control* at lower levels of MSE. Compared to the *Low-Debt* classification, we observe an even larger fraction of debt-averse subjects in the *Borrow-Debt* treatment, in line with our experimental evidence. While it is beyond the current scope of our model to capture this difference, one potential way to rationalize it is by introducing an asymmetric fixed cost to indebtedness, which could also explain the lower average $\hat{\lambda}$ in the *High-Debt* condition. For example, individuals could get an additional debt-free premium when fully repaying their debt. Since in the *High-Debt* condition, outstanding debt can never be fully repaid, the premium is unachievable, reducing the incentives for repaying debt. Conversely, an individual could incur the penalty going from zero to positive outstanding debt, discouraging borrowing even further. Such model will change the interpretation of our current λ as the sum of debt-aversion plus a debt-free premium.³⁸

Overall, our estimation shows the existence of substantial debt aversion toward negative balances. While around one-third to one-half of the subjects seem to maximize returns regardless of their accounts' remaining balances, a significant portion exhibit behavior consistent with debt aversion, ranging from one-quarter in the *Low-Debt* case to almost one-half in the *Borrow-Debt* treatment. Taking our type distribution at face value, we estimate an average for $\hat{\lambda}$, the parameter for debt overweight in our model. Overall, subjects perceive \$1 less in debt as equivalent to \$1.035 in savings, or \$1.065 if we use the *Borrow-Debt* estimates.³⁹ Using the one-shot version at the end of the survey yields an average $\hat{\lambda}$ of 0.025, slightly below that from the *Low-Debt* treatment.

These estimates have sizable implications in terms of investment. An alternative way to interpret λ is by looking at what interest rate people would be willing to borrow to earn a guaranteed 10 percent return. In the case of $\lambda = 0.034$, our estimated weighted average in the *Low-Debt* case, people will not borrow at interest rates higher than 6.38 percent. If we use the average from the *Borrow-Debt* case ($\lambda = 0.067$), the threshold becomes 3.10 percent.

VI. Discussion

A. Understanding Debt Aversion

Across many of our treatments and the structural exercise, we find substantial heterogeneity in debt-biased behavior. While it is not our aim to understand this

³⁷ Some of these participants could be using other non-debt-based heuristics such as equal splitting or a balance matching strategy as in Gathergood et al. (2019) and Ozyilmaz and Zhang (2019).

³⁸ Note that the difference in average $\hat{\lambda}$ between *Low-Debt* and *Borrow-Debt* is 0.03, similar to the difference between *High-Debt* and *Low-Debt* and suggesting that could be the debt-free premium.

³⁹ Both averages are robust to the use of different MSE thresholds. See Figures A.10a and A.10b in the online Appendix for details.

variation in debt attitudes, we consider it relevant for future research to highlight several potential explanations based on the literature.

Previous work has shown how traumatic historical events, such as the Great Depression, can have long-lasting effects on financial decisions (Graham and Narasimhan 2004; Malmendier and Nagel 2011; Laudenbach, Malmendier, and Niessen-Ruenzi 2020) and can even trigger a change in preferences (Bernheim et al. 2021). Similarly, negative experiences with debt, either first- or second-hand, could potentially induce changes in debt aversion. For example, Malmendier, Tate, and Yan (2011) find that CEOs who grew up during the Great Depression are also more averse to debt and lean excessively on internal finance.

Another explanation refers to cultural differences and social norms on perceptions of debt. In many countries, there are strong feelings associated with debt; some have even permeated into language and religious practices. For example, the German word for debt, *schuld*, is the same as the word for guilt. In the United Kingdom, while the English version of the Lord's Prayer says "forgive us our sins as we forgive those who sin against us," the Scottish version says "forgive us our debt, as we forgive our debtors."⁴⁰ Examples of debt-biased behavior are also present in Islamic cultures, where a considerable stigma is associated with borrowing (Adamu et al. 2016). In a recent paper, Azmat and Macdonald (2020) argue that the psychological burden of debt drives the desire to repay home mortgages quickly in Pakistan, at the cost of additional fees and substantial loss in savings for debtors. One alternative explanation for such emphasis on mortgage repayments is the fear of bankruptcy. Ganong and Noel (2023) show how most mortgage defaults in the United States are driven by substantial negative life events rather than when the asset loses value, suggesting a strong aversion to default. Debt aversion might be a natural reaction for people trying to avoid default at all costs.

Finally, there is considerable heterogeneity in other behavioral traits, such as risk aversion (Bruhin, Fehr-Duda, and Epper 2010), appetite for competition (Niederle and Vesterlund 2011), or even loss aversion (Sproul and Michaud 2017), which closely relates to our proposed model of debt aversion. Some of these traits could interact with debt aversion and either mitigate or exacerbate its impacts. Of particular interest is the interaction with present focus, arguably the more predominant trait in the borrowing literature and found to correlate with indebtedness (Meier and Sprenger 2010). This raises the question of under what circumstances we expect debt aversion to be the predominant mechanism instead of present focus. We briefly shed light on this issue next, using two simplified examples based on Berkouwer and Dean (2022) and Allcott et al. (2022).

B. *Present Focus versus Debt Aversion*

Using a simple calibration based on previous works, we illustrate how debt aversion and present focus interact. Our main insight is that debt aversion is more likely to be predominant when benefits materialize *after* debt repayments accrue.

⁴⁰ Sources: <https://www.churchofengland.org/> and <https://www.churchofscotland.org.uk/>

TABLE 8—EXAMPLES OF BORROWING DECISIONS

	Scenario	Present focus & time discounting	Debt aversion	Percentage return
<i>Panel A. Example 1: charcoal cookstove investment</i>				
1.	$\beta = 1, \delta = 1, \lambda = 0$	No	No	291
2.	$\beta = 0.80, \delta = 0.95, \lambda = 0$	Yes	No	259
3.	$\beta = 1, \delta = 1, \lambda = 0.15$	No	Yes	253
4.	$\beta = 0.80, \delta = 0.95, \lambda = 0.15$	Yes	Yes	225
5.	$\beta = 0.80, \delta = 0.95, \lambda = 1$	Yes	Yes	148
<i>Panel B. Example 2: payday lending with exponential utility</i>				
1.	$\beta = 1, \delta = 1, \lambda = 0$	No	No	94
2.	$\beta = 0.80, \delta = 1, \lambda = 0$	Yes	No	117
3.	$\beta = 1, \delta = 1, \lambda = 0.15$	No	Yes	88
4.	$\beta = 0.80, \delta = 1, \lambda = 0.15$	Yes	Yes	111
5.	$\beta = 0.80, \delta = 1, \lambda = 1$	Yes	Yes	95

Notes: Panel A simulates borrowing \$40 to purchase a charcoal cookstove using a three-month loan at 14 percent, as in Berkouwer and Dean (2022). Percentage returns use their charcoal savings estimates without incorporating any additional health or environmental benefits. Panel B simulates a two-week payday loan of \$200 to finance an immediate expenditure/consumption at a 15 percent interest rate. Present focus parameters β and δ are chosen based on Allcott et al. (2022) estimates. In addition, for panel B, we assume an exponential utility function with $\alpha = 0.0064$ following their main specification. Results are qualitatively the same with a linear utility function, reported in online Appendix A.21. In both examples, a percentage return higher than 100 percent implies the acceptance of the loan, while a percentage lower than 100 percent implies a loss and the decline of the loan offer.

However, when they accrue before, present focus dominates all but very high levels of debt aversion.

Let us first consider a simple example where an agent borrows a one-month loan amount of x to finance an immediate consumption. With a linear utility function, utility is given by $x + \beta\delta(1+i)(1+\lambda)(-x)$, where i is the loan interest rate. A simple calibration based on Allcott et al. (2022) ($\beta = 0.80, \delta = 1, i = 15$ percent) reveals that you need debt-aversion λ to be much larger than 1.15 for the agent to be unwilling to borrow. Note that this threshold corresponds to the maximum level of debt aversion that we can quantify in our experiment, and that only 14 percent of participants can potentially have those levels in our structural exercise. Hence, it seems unlikely that debt aversion will play a predominant role when borrowing finances present consumption.

Consider now an agent that borrows to finance a long-term investment. To simplify, assume a one-month loan and an investment with a guaranteed return of r one-month after investing. Thus, benefits can repay the loan, and liquidity constraints play no role. Using a linear utility function, we have $(1+i)(1+\lambda)(-x) + (1+r)x$, and since repayment and benefits accrue simultaneously, discounting terms drop out of the expression. Without debt aversion ($\lambda = 0$), an agent will borrow as long as $r > i$. However, with debt aversion ($\lambda > 0$), borrowing occurs only if $(1+r) > (1+i)(1+\lambda)$. As discussed in Section V, a coefficient of debt aversion $\lambda = 0.035$ is enough to deter a 10 percent guaranteed return if the interest rate is no lower than 6 percent, a 4-point difference. In cases with different timing, present focus interacts with debt aversion, making the comparison less straightforward. Nonetheless, the main intuition still holds: it is in long-term investments where we expect debt aversion to matter the most.

Building on these intuitions, we perform two calibration exercises based on Berkouwer and Dean (2022) and Allcott et al. (2022) reported in Table 8. The first portrays a long-term investment, a charcoal cookstove that halves daily consumption, that requires a high upfront cost, and where borrowing costs are due before the agent can reap all the benefits. Panel A shows that a debt aversion λ of 0.15⁴¹ alone reduces the return on the investment as much as present focus. Combining both mechanisms further reduces returns, although one would need very high values of debt aversion to deter investment.⁴² In this situation, since benefits materialize over a longer period than debt repayments, time discounting compounds the impact of debt aversion.

Inspired by Allcott et al. (2022), the second exercise considers a two-week loan to finance immediate consumption at a 15 percent interest rate.⁴³ Panel B illustrates that when the benefits of borrowing materialize earlier than its costs, it is unlikely that we find debt aversion as a dominant force. Future loan repayments are heavily discounted by the interaction with present focus, counteracting the overweight caused by debt aversion. A present focus agent will accept the payday loan even for a debt aversion λ of 0.15. Only at high levels of debt aversion does the agent decline the loan.

While at first, debt aversion might be in tension with the idea that people are present focus and borrow too much, this is not necessarily the case. Our simple calibration highlights why most empirical evidence on borrowing and present focus comes from settings where borrowing funnels present consumption.⁴⁴ However, as panel A illustrates, debt aversion might play a significant role in long-term decisions and, depending on the timing of disbursements, might even compound with present focus. This might explain why there is documented reluctance to borrow for investing.⁴⁵

VII. Conclusion

Debt aversion provides a different scope for assessing debt policies' unintended benefits and consequences. Relief and debt forgiveness programs could provide additional benefits beyond their direct wealth impact, as they would also lessen the detrimental effects of debt aversion in subsequent financial decisions. Along these lines, Ong, Theseira, and Ng (2019) documents improved cognitive capacity and reduced anxiety from a debt-alleviation policy, and Duquenois and Jagnani (n.d.) finds improved sleep quality. In some cases, reducing debt could create a virtuous cycle, where lower debt improves subsequent financial decisions and prevents indebtedness in the future. On the other hand, if indebtedness induces suboptimal decisions, even short-term borrowing in times of financial hardship could have

⁴¹ This value approximately corresponds to the high λ types from Section V.

⁴² While we cannot rule out the possibility of such high values, it is likely that other mechanisms like risk attitudes or liquidity constraints also play a role in this scenario.

⁴³ To keep things simple, we rule out the possibility of revolving the payday loan.

⁴⁴ Karlan, Mullainathan, and Roth (2019); Allcott et al. (2022) are examples of short-term borrowing at high-interest.

⁴⁵ In investment decisions with long time horizons, such as a charcoal cookstove (Berkouwer and Dean 2022), a business micro-loan (Liu and Roth 2022), or college education (Callender and Jackson 2005), people seem reluctant to borrow.

long-term consequences. Gelman et al. (2015) show that a slight shift in income timing can drive extended indebtedness for highly constrained consumers. Similarly, Leary and Wang (2016) show that suboptimal savings and consumption decisions cause a significant fraction of the demand for payday loans, and Carvalho, Olafsson, and Silverman (2019) find that “misfortunes” increase demand for payday loans.

A possible way to alleviate the detrimental consequences of debt aversion is by changing how we present borrowing opportunities and design repayment schemes. For example, financial aid programs in the form of grants might be more effective than loans in encouraging uptake. Prospective college students might be more likely to enroll after receiving a grant than a loan, even if the return on higher education compensates for the amount borrowed. Field (2009) provides evidence of such framing effects by showing that students who receive a grant instead of a financially equivalent student loan are more likely to pursue a career in the public sector.

In our setting, many try to rid themselves of their debt as quickly as possible. However, this does not necessarily have to come at the expense of maximizing earnings in other settings. Alternative debt repayment methods like income-based repayment could mitigate the detrimental consequences of debt aversion by better aligning the incentives for maximizing earnings and repaying debt faster. Policies allowing for deferring debt repayment, such as those for federal student loans, could also be beneficial. Along these lines, Hershfield et al. (2015) advocates incorporating behavioral and psychological biases into the policy discussion around indebtedness. One challenge of incorporating debt aversion as a behavioral bias in policy is that it took a week-long experiment to measure it in our context. However, the one-shot version of our design, highly correlated with the week-long responses, can be easily incorporated into surveys and field work to elicit debt aversion. To capture the full repercussions of debt, we must be able to measure and understand all of its consequences.

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